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Course 66

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Predictive Antimicrobial Stewardship: A Computational ARIMA Approach to Forecast *Salmonella enterica* serovar Typhi Resistance Dynamics in Urban Bangladesh

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1. Introduction

14 Typhoid fever, caused by the bacterium *Salmonella enterica* serovar Typhi, is a massive and escalating public health crisis across South Asia [1, 14]. In Bangladesh, the disease spreads even faster due to our dense urban population, struggling water sanitation infrastructure, and the heavy floods we get every monsoon season [2]. Historically, clinics treated typhoid easily with older antibiotics like chloramphenicol, ampicillin, and trimethoprim sulfamethoxazole. However, the overavailability of these drugs created intense evolutionary pressure, making them clinically useless and giving rise to multidrug resistant (MDR) strains [3].

To fight these MDR strains, local clinics shifted their focus to fluoroquinolones (mostly ciprofloxacin) and macrolides (like azithromycin) [4]. Unfortunately, this caused a secondary crisis. Recent longitudinal surveillance in Dhaka shows that ciprofloxacin efficacy has plummeted, with about 95% of typhoid strains now showing resistance [5]. Oddly enough, as we stopped relying on the older drugs, resistance to those original treatments has paradoxically decreased. The problem is that our current antimicrobial resistance (AMR) surveillance relies almost entirely on retrospective data [6]. We usually only document resistance after a patient treatment fails. To preserve the medicines we have left, we need to transition our surveillance from a simple reporting model to a predictive computational framework.

2. Theoretical Framework and Literature Review

16 Global tracking for antimicrobial resistance is mostly handled by the WHO Global Antimicrobial Resistance and Use Surveillance System (GLASS) [7]. This database is excellent for big-picture data on antibiotic consumption. However, international registries move slowly and lack the micro level clinical details hospitals need to make immediate interventions.

In the past, researchers often tried to model this resistance using standard linear regression. But linear models do not work well for infectious diseases. They completely miss the chaotic seasonal outbreaks that happen after the monsoons, and they cannot process the complex ways bacteria share resistance genes. **Applying systematic evidence synthesis methodologies**, it became clear that just looking back at historical data is not enough. To fix these mathematical
18 limitations, this study proposes using an ARIMA (AutoRegressive Integrated Moving Average) time-series model [8]. ARIMA is a standard epidemiological tool designed to handle messy data by tracking past trends and smoothing out random environmental shocks. By combining the
1 broad consumption data from WHO GLASS with the local clinical archives from the Bangladesh Directorate General of Health Services (DGHS) [9], we can actually calculate how our local prescribing habits exert selective pressure and drive resistant strains.

3. Research Objectives and Hypotheses

This study has three main objectives:

1. Find the link: Calculate the correlation between the use of fluoroquinolones and macrolides and the shifts in *S. Typhi* resistance in Bangladesh over the last ten years.
2. Future forecasting: Train an ARIMA model to project the resistance path of *S. Typhi* for the next 36 months [8].
3. Clinical action: Turn these computational predictions into a "Local Stewardship Blueprint" that gives hospital managers a clear timeline to rotate antibiotics before the current ones fail completely [10].

Hypotheses:

- H1: The statistical analysis will show a delayed but positive correlation between the massive spike in azithromycin use and new resistance markers in local blood and stool cultures.
- H2: The 36-month ARIMA forecast will show that ciprofloxacin resistance will remain above 85%, proving it is no longer a reliable treatment even if the government restricts its distribution [5].

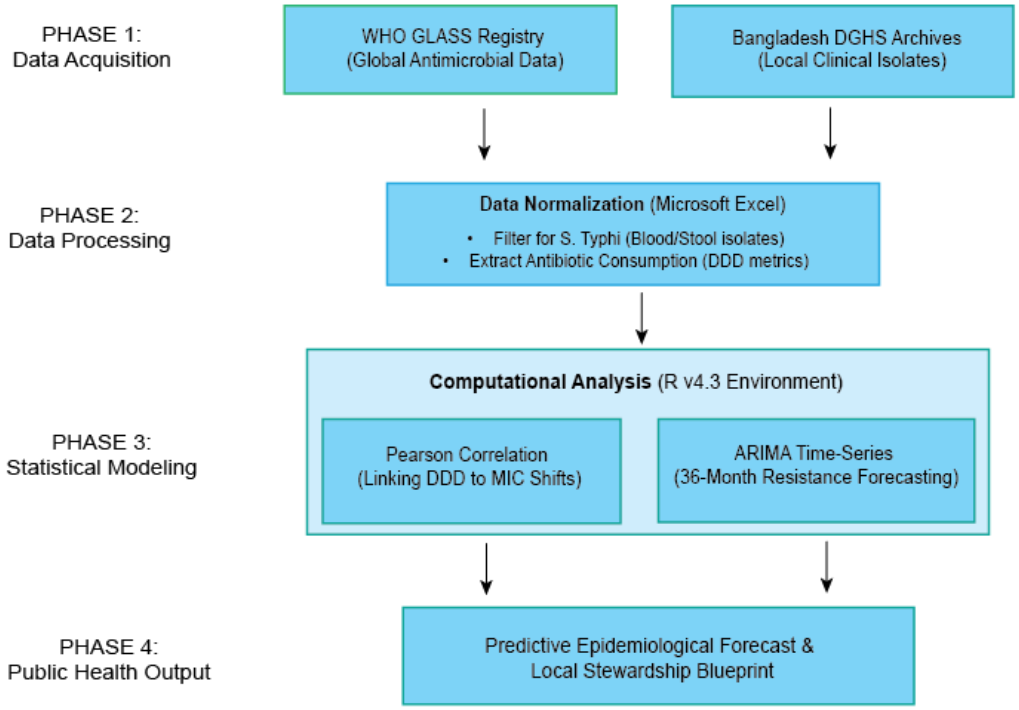
4. Methodology and Computational Pipeline

4.1 Phase 1: Data Acquisition

First, I will extract the systemic antibiotic consumption numbers from the WHO GLASS registry, specifically focusing on South Asia [7]. Next, I need the localized resistance profiles, which I will pull from the DGHS archives [9]. I am strictly looking for confirmed blood and stool cultures that tested positive for *S. Typhi* across major hospitals in the Dhaka district.

4.2 Phase 2: Data Normalization

Raw hospital datasets usually have missing files and formatting errors. I will use Microsoft Excel to clean the data first. If patient data points are missing, I will use kNN (k-nearest neighbors) imputation to fill the gaps without distorting the real statistics [11]. I also need to convert the antibiotic consumption amounts into Defined Daily Doses (DDDs) to match standard WHO protocols [7]. Finally, the raw bacterial MIC values will be log-transformed so the variance does not interfere with the R algorithms.



4.3 Phase 3: Computational Modeling

I will run the statistical modeling in R (version 4.3). First, a Pearson correlation matrix will link the DDD consumption data to the MIC shifts, which mathematically shows how local drug use drives the selection of resistant phenotypes. Next, I will build the ARIMA forecasting model [8]. After running an Augmented Dickey-Fuller test to ensure the data is stationary, I will use the auto.arima() function to find the best hyperparameters. The final model will run a 36 month simulation to map out when our current treatments will likely fail, while making sure to factor in the huge seasonal infection spikes we see during the monsoon [2].

5. Ethical Considerations and Feasibility

Working with live, drug resistant *S. Typhi* requires a Biosafety Level 2 lab, which is not feasible for a high school research project. Because I am running a purely computational study using public open-access data, I completely avoid any biohazard risks while still tackling a major local problem. This also means IRB approval is not required since there are no direct human subjects involved.

This setup is highly modular. Once the R pipeline works, other students could easily swap the typhoid parameters for datasets on diseases like cholera. It shows that students in developing countries can still do high level disease research without access to expensive physical labs.

6. The 12 Week Execution Timeline

To make sure the project stays on track, I would set a strict 12 week schedule:

- Weeks 1 to 3: Download the WHO GLASS and DGHS datasets, clean the CSV files, and check for formatting errors.
- Weeks 4 to 6: Run the kNN data imputation in Excel, convert antibiotic consumption into DDDs, and log transform the bacterial MIC values [11].
- Weeks 7 to 9: Set up the R environment, run the Dickey Fuller tests, and tune the ARIMA model.
- Weeks 10 to 12: Generate the final 36 month forecasting graphs and write out the Local Stewardship Blueprint.

7. Limitations

Even with strong data, this method has some limitations. Relying on the DGHS archives means we are mostly looking at patient data from large, centralized hospitals in Dhaka [9]. This creates an urban reporting bias, leaving rural resistance profiles unclear until we can conduct physical surveys later. Also, ARIMA modeling assumes that future trends will follow past patterns. If the government suddenly changes its healthcare policies, the algorithm will need to be adjusted. Finally, these human focused databases completely miss agricultural antibiotic runoff from poultry farms, which is a massive hidden cause of resistance in our local water supply [12].

8. Expected Outcomes and Public Health Impact:

Good research needs a measurable, real world impact. The main goal here is to build a tool that local clinics can actually use.

If ciprofloxacin fails completely, clinics will have to rely on IV ceftriaxone. That requires hospital beds, sterile IV needles, and trained nurses, resources that run out incredibly fast during the outbreak seasons [13]. By predicting exactly when our current drugs will hit a critical failure point, hospital administrators can proactively rotate their antibiotic stocks [10]. Moving to a predictive defense model like this will save our backup drugs, optimize hospital budgets, and most importantly, save lives when the next typhoid wave hits.

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